1 Background and Relevant Research to Date

Background

My upbringing and personal history have instilled in me a lifelong appreciation for bringing reason into public policy, along with an aspiration for compassionate policy debates where we all start from a common understanding of the evidence. Within this framework, I have a particular interest in investigating how societal support helps people overcome adversity and thrive. Figure 1 shows my prescriptive view of how to make progress toward these goals through research and policy analysis.

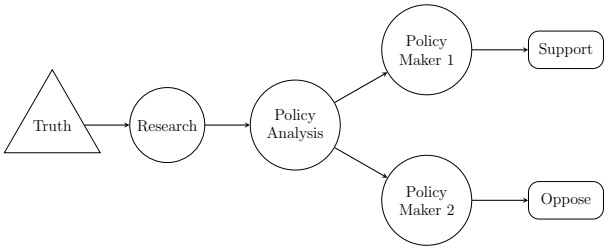


Figure 1: The evidence-to-policy chain.

In my work to date, I have used tools from policy analysis, economics, and research transparency to conduct research into how evidence of the impacts of previous policies is reported and aggregated across studies in order to clearly establish where the scientific consensus is and where it is not. I combine this research with fully appraisable policy analyses that use existing evidence to describe the costs and benefits of prospective policies. With agreed-upon costs and benefits, different policymakers can reveal their normative views by choosing to support or oppose those policies (as shown in Figure 1). When research and policy analysis fail to build that common ground, we end up contributing to a world of alternative facts, or what Manski described as *dueling certitudes* (Manski, 2013).

Before my PhD, I worked at Chile's Ministry of Planning, where I produced official poverty estimates and co-led the evaluation of Chile Solidario, the national anti-poverty program. I also worked at the Chilean Ministry of Finance, where I quantified the budget impact of a range of social policies. These poverty and other policy impact estimates reached decision makers as single figures with no consideration of uncertainty or ability to inspect the findings, an experience which first pushed me to think more about transparency in policy analysis.

As a result, my doctoral and early post-doctoral work sought to increase transparency at the policy analysis node of the Figure 1 chain. My PhD dissertation took a Congressional Budget Office (CBO) estimate of the employment effects of a minimum wage increase and rebuilt it as a fully transparent and reproducible analysis, so that a reader could trace every input and see how the result moved under alternative assumptions (Hoces de la Guardia, 2017). For my post-doctoral work at the University of California, Berkeley, I turned the dissertation's case study into a general framework for designing transparent and reproducible policy analyses, called Open Policy Analysis. I also trained researchers in research transparency and developed tools and guidance for computational reproducibility (Brodeur et al., 2025, 2026).

In 2023 I was given the opportunity, for the first time in my career, to work full time on the research node of the Figure 1 chain. I led a multi-year project on the standardization of results in economics, in collaboration with a team of senior researchers that includes Edward Miguel, Erik Sørensen, and Bertil Tungodden. The project's first application, and the subject of my job market paper, is the under-reporting of findings from pre-registered randomized controlled trials (RCTs). The rest of this statement describes this project and how it connects with my transparency in policy analysis work to form two complementary lines of research with one aim: to develop and test innovations that strengthen the evidence-to-policy link.

Current and Previous Research

As described above, my research focuses on investigating whether and how transparency can increase our agreement on which evidence and which policy analyses are the most credible. My work has focused on two nodes of the chain in Figure 1. At the research node, I investigate in my job market paper whether findings become public, and test new ways to standardize how they are reported. At the policy analysis node, I developed Open Policy Analysis to make the data, code, and assumptions behind a policy estimate easily inspectable.

Job Market Paper: Measuring and Recovering Missing Results
My job market paper, "Beyond Publication Bias: Characterizing and Understanding Missing Results in Economics" (with Miguel, Sørensen, Tungodden, et al.), asks how much of what economists plan to test is never reported, and why. Using the American Economic Association (AEA) RCT Registry, where researchers record their hypotheses before collecting data, we developed a method to standardize each pre-registered hypothesis and track it to its reported result. We also conducted an RCT with research teams to test plausible barriers to results reporting. I am co-principal investigator (co-PI) of the program behind the paper, funded by a three-year grant from the Centre for Experimental Research on Fairness, Inequality and Rationality (FAIR) at NHH Norwegian School of Economics. I led the design and implementation of the RCT and the hypothesis-encoding protocol, managing a team of three research assistants.

Across 5,057 pre-registered hypotheses from 317 studies, 58% (on average across studies) have no publicly available results 8 to 9 years after pre-registration. Among hypotheses with reported results, a large share, 64.5%, are null, which cuts against the common narrative that only statistically significant findings become publicly available.

To learn what keeps results unreported, we ran an RCT with over 300 research teams, testing three plausible barriers: a lack of *awareness*, of *engagement*, and of *resources*. Only the engagement intervention, in which our team identified each author's specific missing hypotheses and asked them, one by one, to report or explain each one, significantly increased reporting. Neither a light-touch awareness message nor a more expensive research-assistance intervention had an impact. Engagement raised the share of hypotheses with publicly available results by 6.2 percentage points (11% of missing hypotheses). Most of the gain came from results that existed but were buried. Genuinely new estimates account for 2.5 of the 6.2 percentage points. Authors also explained a further 18% of missing hypotheses (about 10% of all hypotheses) as research never completed, abandoned at varying points before final data collection. Even with engagement, 42% of all hypotheses remained neither reported nor explained. Recovery after the fact is possible but labor-intensive. More than a decade after the registry opened, stronger incentives and infrastructure are still needed to make sure researchers report the evidence they planned to generate, rather than relying on others to recover it later. The paper prototypes one piece of possible infrastructure, standardized reporting templates we call Results Reports. These reports are fixed-format records stating each hypothesis, population, and outcome definition alongside its result.

Open Policy Analysis: Feasible, but (Previously) Costly to Implement
Open Policy Analysis (OPA), published in *Science and Public Policy* (Hoces de la Guardia et al., 2021), seeks to translate research transparency practices into policy analysis. Every policy estimate combines three kinds of input (research, data, guesswork), and the policy analyst's choices about each of them compound, so the final number carries degrees of freedom a reader cannot see. The OPA framework sets out three principles (open output, open analysis, and open materials) and applies them to four layers of a report, each written for a different reader: from the policymaker, who sees a single pre-specified result and how it moves with the key assumptions,

down to the analyst, who can reproduce everything. We tested the OPA framework in policy analyses of a progressive wealth tax, with Emmanuel Saez and Gabriel Zucman (Hoces de la Guardia et al., 2019), and of deworming interventions, with Edward Miguel (Hoces de la Guardia and Miguel, 2021). I applied it in a transparency assessment of ex-ante economic analyses at the U.S. Agency for International Development (USAID; Hoces de la Guardia, 2022), and presented it, with tailored recommendations, to the CBO and Chile's Ministry of Finance. I also co-authored guidance on transparent research that the Inter-American Development Bank (IDB) adopted for its research staff (Hoces de la Guardia and Sturdy, 2018).

The main lesson is that open analysis is feasible but costly. Teams and policy audiences engaged, yet each application was built by hand, over several months of work, and none became routine practice. My working hypothesis is that tools built on large language models (LLMs), including agentic ones that carry out multi-step tasks, can lower that cost enough to reopen the agenda. I am currently prototyping two additional open analyses, each built in weeks rather than months, available on my personal website (`fhoces.github.io/policy.html`).

2 Directions for Future Research

Two Complementary Lines of Research

My main research line going forward is the standardization of research results, which builds on my job market paper and on a well-developed area of research in medicine and public health, where it is known as reporting guidelines (e.g., CONSORT for randomized trials; Schulz et al., 2010). OPA complements this primary research line. Both address one problem: when the quality of evidence or analysis cannot be observed, audiences may judge it by which result best fits their normative views, or by poor proxies such as the presenter's fluency, confidence, and credentials. By recording hypotheses, populations, and other methodological choices in a structured format, Results Reports make it easier to see what was and was not reported, and to understand when studies of the same question can be compared and aggregated, and why they disagree when they do. OPA makes it harder to pick empirical results to fit a normative position and replaces those proxies with the ability to appraise a policy analysis on its methods alone. Each line of research makes the other more valuable: an open cost-benefit analysis is more useful when the literature it draws on reports the results it planned to report, and standardized results are more useful when the analyses that use them can be thoroughly inspected.

Line 1: Standardization of Research Results
Short-term projects. The first of these projects is already in progress and three more are feasible with the data in hand.

- An LLM-based tool to standardize pre-registered hypotheses and their results, benchmarked for accuracy and cost against our hand-encoded dataset. This project is ongoing, funded by a one-year grant from Coefficient Giving, on which I am co-PI. Encoding is the costly step behind engagement because, before engaging an author, we must standardize every hypothesis and locate each reported result. If the tool reliably matches hand encoding at a fraction of the cost, engaging authors becomes affordable across the registry.
- Testing for spillovers from our interventions to the networks of researchers in the treatment arms. These tests are pre-registered and reported briefly in my job market paper, and a full paper would develop them even further.
- A multiple-testing correction for every study in the sample, using the true number of hypotheses it pre-registered rather than the number it reported.

- A test of publication-bias corrections against directly measured missing results. We would test whether a standard model-based correction (e.g., Andrews and Kasy, 2019) for one or a few literatures in our sample (such as cash transfers) matches the results we observe.

Medium-term projects. Two projects build on the short-term projects described above.

- A second RCT on research teams, now run at the moment of RCT registration. This RCT would test whether adopting Results Reports as a standard raises the share of pre-registered hypotheses reported. My collaborator Edward Miguel helps oversee the AEA RCT Registry and can pilot it there. The project would retest the awareness message by sending it early and by the adopting institution rather than by our team. It would also test engagement at scale, if the LLM tool makes engagement affordable.
- Results Reports for quasi-experimental and descriptive research. For quasi-experimental methods such as instrumental variables, regression discontinuity, difference-in-differences, and, more recently, movers designs built on two-way fixed-effects models, Results Reports would ask authors to state the hypothetical experiment each design seeks to approximate, showing when results can be aggregated and when they cannot. For descriptive studies that measure the same quantity and disagree sharply, Results Reports would show where the differences lie and how much each methodological choice contributes to them, as in estimates of U.S. income inequality (Piketty et al., 2024; Auten and Splinter, 2024).

Line 2: Open Policy Analysis
I plan to build more open policy analyses, exploring what LLM tools can do at each layer of an analysis and turning the lessons into guidance for policy analysts. A key collaborator in this effort is Chapin White (University of Southern California). We have a proposal under review to analyze policy options for a federal program that discounts prescription drugs for hospitals and clinics serving low-income patients (the 340B Drug Pricing Program). We are also exploring other areas of collaboration. Additional case studies could build on earlier work with UC Berkeley colleagues including Emmanuel Saez on taxation and David Anthoff, with whom I developed a proposal on the economics of climate change.

OPA also combines with the standardization of results in one further project: identifying, before the research is done, where the next study would be most informative. An open analysis shows which inputs move a policy estimate most; standardized results show how much evidence stands behind each input and how much it disagrees. Together they point to the input where the marginal study is most likely to change the answer.

This second line is more exploratory and riskier than the first. Its products are open policy analyses and guidance for analysts rather than papers, so its output is less straightforward to turn into peer-reviewed publications. The downside is nonetheless limited: if it succeeds, it adds a second stream of publications; if it does not, it still contributes to policy practice and to teaching.

I also have close collaborators at Universidad de Chile, Pontificia Universidad Católica de Chile, and Chile's Ministry of Finance. No project is under way with them yet. I plan to start joint projects with them in Latin America on both lines of research. I will pursue both lines with dedication, and I see them as a starting point for new ideas with future colleagues and with these partners.

References

Isaiah Andrews and Maximilian Kasy. Identification of and correction for publication bias. *American Economic Review*, 109(8):2766–2794, August 2019. doi: 10.1257/aer.20180310. URL https://www.aeaweb.org/articles?id=10.1257/aer.20180310.

Gerald Auten and David Splinter. Income inequality in the United States: Using tax data to measure long-term trends. *Journal of Political Economy*, 132(7):2179–2227, 2024.

Abel Brodeur, Seung Yong Sung, Edward Miguel, Lars Vilhuber, and Fernando Hoces de la Guardia. Assessing reproducibility in economics using standardized crowd-sourced analysis. Working Paper 33753, National Bureau of Economic Research, May 2025. URL https://www.nber.org/papers/w33753.

Abel Brodeur, Derek Mikola, Nikolai Cook, Lenka Fiala, ..., Fernando Hoces de la Guardia, et al. Reproducibility and robustness of economics and political science research. *Nature*, 652: 151–156, 2026. doi: 10.1038/s41586-026-10251-x.

Fernando Hoces de la Guardia. *How Transparency and Reproducibility Can Increase Credibility in Policy Analysis: A Case Study of the Minimum Wage Policy Estimate*. PhD thesis, Pardee RAND Graduate School, 2017. URL https://doi.org/10.7249/RGSD394.

Fernando Hoces de la Guardia. Transparency assessment of ex-ante economic analyses in development organizations. Technical report, USAID, November 2022. URL https://web.archive.org/web/20250202082838/https://pdf.usaid.gov/pdf_docs/PA00ZQ1N.pdf.

Fernando Hoces de la Guardia and Edward Miguel. Open Policy Analysis of Deworming Interventions. Technical report, Berkeley Initiative for Transparency in the Social Sciences (BITSS), April 2021. URL https://www.bitss.org/research/opa/projects/deworming/.

Fernando Hoces de la Guardia and Jennifer Sturdy. Best practices for transparent, reproducible, and ethical research. Technical Note 1635, Inter-American Development Bank, March 2018. URL https://doi.org/10.18235/0001564.

Fernando Hoces de la Guardia, Emmanuel Saez, Sara Stoudt, and Gabriel Zucman. Open Policy Analysis for a Progressive Wealth Tax. Technical report, Berkeley Initiative for Transparency in the Social Sciences (BITSS), March 2019. URL https://www.bitss.org/research/opa/projects/progressive-wealth-tax/.

Fernando Hoces de la Guardia, Sean Grant, and Edward Miguel. A framework for open policy analysis. *Science and Public Policy*, 48(2):154–163, June 2021. ISSN 0302-3427, 1471-5430. doi: 10.1093/scipol/scaa067. URL https://academic.oup.com/spp/article/48/2/154/6018524.

Charles F. Manski. *Public policy in an uncertain world: analysis and decisions*. Harvard University Press, 2013. ISBN 0-674-06754-1.

Thomas Piketty, Emmanuel Saez, and Gabriel Zucman. Income inequality in the United States: A comment. Working paper, Paris School of Economics and UC Berkeley, September 2024. URL https://gabriel-zucman.eu/files/PSZ2024.pdf.

Kenneth F. Schulz, Douglas G. Altman, and David Moher. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. *Journal of Clinical Epidemiology*, 63 (8):834–840, 2010.